
course: ml-batch-1

audience: student

kind: project

id: p01

title: "Project 1 — Exploratory Data Analysis on a Real Dataset"

issued: "Week 1, Day 5"

due: "Week 2, Day 5"

est_hours: 5

weight_pct: 1.5

difficulty: "Easy Medium"

Project 1 — EDA on a Real Dataset

Issued W1 D5 · due W2 D5 · ~5 hours · 1.5% of your grade

The brief

Choose a real dataset, perform a complete exploratory data analysis, and produce a notebook plus a written findings document.

You are not building a model. You are finding out whether a model would be worth building, and what would go wrong if someone tried.

Why this project exists

Every project after this one begins with the work you are about to do, and the quality of it determines everything downstream. A model trained on data whose sentinels were never spotted is wrong in a way no amount of tuning will fix.

It also builds the habit this course cares most about: **stating findings, not observations**. The marks here are weighted heavily toward the written document, not the notebook.

Choosing a dataset

Requirements:

- **Real**, not generated. Someone collected it about something.
- **At least 500 rows and at least 6 columns**, with a mix of numeric and categorical.
- **Openly licensed**, and you state the source and licence.
- **Messy**. A perfectly clean dataset gives you nothing to find. If your chosen dataset has no missing values and no oddities, pick a different one.

Good starting points:

Dataset	Where	Why it works
Titanic	Kaggle / seaborn	Missing ages, mixed types, obvious target
Ames Housing	Kaggle	80 columns, real missingness patterns, regression target
Adult / Census Income	UCI	Categorical-heavy, ? sentinels, class imbalance
Telco Customer Churn	Kaggle	Business framing, a numeric column stored as text
NYC Airbnb listings	Kaggle	Skewed prices, geography, text fields
Any dataset from your own work or field	—	Best option if you have one

If you use data from your own workplace, **check you are allowed to** and remove anything identifying before it goes near a public repository.

Requirements

R1 — Dataset overview (must have)

- Shape, column names, dtypes.
- **A sentence stating what one row represents**, and a check that the data agrees (uniqueness of any id column).
- Where the data came from, when it was collected, and its licence.

R2 — Data quality audit (must have)

Find and report every one of these that applies:

- Missing values, per column, as counts and percentages.
- Duplicated rows and duplicated ids.
- Impossible values (negative ages, future dates, percentages above 100).
- Sentinel values pretending to be data (999 , -1 , ? , N/A , 0 where zero is impossible).
- Columns with the wrong dtype (a number stored as text, a date stored as text).
- Constant or near-constant columns.

Do not impute anything. Record what is wrong and what you would do. Imputing during EDA is a specific mistake this project is checking for.

R3 — Target analysis (must have)

Pick a target variable — the thing a model would predict — and state why it is a sensible target.

- Distribution: histogram, mean, median, standard deviation, min, max.
- Skew: is the mean far from the median, and what would you do about it?
- For classification: the class balance, as percentages.
- Outliers: how many, how extreme, and are they errors or real?

R4 — Relationships (must have)

- Correlation matrix of numeric features, with a heatmap.
- Target against each of at least four features — scatter for numeric, boxplot for categorical.
- **At least one feature where the correlation is misleading** — either near zero with a visible non-linear relationship, or high because of an obvious confounder. If your dataset genuinely has none, say so and show the evidence.

R5 — The findings document (must have — highest weighted section)

A markdown document, in the notebook or alongside it, with numbered findings in this form:

```
N. [What is true] [why it matters] [what you would do].
```

Structure it as: Dataset, Data Quality, Target, Relationships, **Risks**, Next Steps. The Risks section must contain at least three entries — things that would make a model built on this data misleading. Examples: leakage candidates, no timestamp so no temporal validation, a rare category that a random split could strand, a target defined in terms of a feature.

Minimum eight findings. Quality over quantity: three excellent findings and five weak ones scores worse than eight solid ones.

R6 — Reproducibility (must have)

- The notebook runs top to bottom on a fresh kernel without error.
- Every random operation is seeded.
- The data is loaded from a URL or a documented path, not from a file only you have.

R7 — Stretch (optional)

- A `missingno` -style visualisation of missingness patterns, and an analysis of whether missingness in one column correlates with values in another.
- A feature you engineered from existing columns, with an argument for why it should help.
- A comparison of the same relationship measured by Pearson and Spearman correlation, with an explanation of when they disagree.

What to submit

```
week01/  
  project1_eda.ipynb    the analysis
```

FINDINGS.md	the written document (may also be in the notebook)
README.md	dataset source, licence, how to run

Push to your course repository and post the link.

Marking — 100 points

Area	Points	Full marks means
Findings document	30	8+ findings, each with implication and action; 3+ genuine risks
Data quality audit	20	Every problem in the data found, including the subtle ones
Target analysis	15	Distribution, skew, balance, outliers — all addressed
Relationships	15	Correct plots, correct reading, the misleading-correlation case
Plot quality	10	Labelled axes, titles that state findings, readable
Reproducibility & code	10	Runs clean top to bottom, seeded, documented source

Bonus up to +5 for R7. Capped at 100.

Automatic deductions

Issue	Deduction
Notebook does not run top to bottom	−15
Imputation performed during EDA	−10
Any plot without axis labels	−2 each, max −10
No dataset source or licence stated	−5
Findings that are observations only (“age has missing values”)	−2 each
Late without an extension	−30% of earned marks

How to approach this

Session 1 (~1h). Choose the dataset. Load it. Run `shape`, `head`, `info`, `describe`, `isna().sum()`. Write down, in plain text, three things that already look wrong. Stop there.

Session 2 (~1.5h). The quality audit. Go through R2 item by item. This is detective work, not plotting — most of it is `value_counts()`, `describe()` and checking whether the extremes make sense.

Session 3 (~1.5h). Target and relationships. Plots, and a written sentence under each one before moving on. **Write the sentence at the time**; going back later to interpret twenty plots is miserable and it shows.

Session 4 (~1h). Assemble the findings document, restart the kernel, run all, fix what breaks, write the README, push.

Where people lose marks

- **The findings document written last, in fifteen minutes.** It is 30% of the grade. Write it as you go.
- **Observations instead of findings.** “There are 177 missing ages” is worth nothing on its own.
- **Twenty plots, no interpretation.**
- **Missing the sentinel.** Everyone finds the `NaN` s. The mark is in finding the `-1`, the `999`, the `?`, and the suspicious spike at zero.
- **A notebook that does not re-run.** Restart the kernel and run all, before submitting. Every time.
- **Claiming causation.** “Being in first class caused survival” is not something this data can support. “First-class passengers survived at 63% against 24% in third” is.

A note on honesty

If your dataset turns out to be boring — clean, small, with no interesting structure — **say so, in the findings, with evidence.** “This dataset has no missing values, no duplicates, and no feature correlating with the target above 0.15; a model here would be unlikely to beat the mean” is a legitimate and well-marked conclusion.

Inventing significance you did not find is the one thing that will cost you more than finding nothing.